

SIO-UV: Rapid and Robust Sonar Inertial Odometry for Underwater Vehicles

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Abstract—We propose a sonar-inertial odometry framework named sonar-inertial odometry for underwater vehicles (SIO-UV) for underwater vehicles, designed to address the challenges posed by complex underwater environments and sensor limitations. The aim is to achieve efficient and robust simultaneous localization and mapping (SLAM) in underwater conditions. SIO-UV integrates observation data from an inertial measurement unit (IMU) and a forward-looking multi-beam sonar (FLS) to construct factor graph nodes, using IMU preintegration to estimate the trajectory of the underwater vehicle and to provide an initial pose optimization for sonar observations. To enhance the robustness of sonar data, this method is based on the FLS model and incorporates a multiscale constant false alarm rate (MCFAR) denoising algorithm. It can convert 2-D sonar images into 3-D point cloud data. We introduce a reverse regression mapping method for FLS. Ultimately, the generated sonar point cloud is used to estimate sonar odometry and provide bias correction for IMU preintegration. Extensive validation of underwater mapping was conducted in advanced simulation environments, with quantitative experimental results demonstrating the high accuracy of SIO-UV in underwater vehicle odometry estimation. Furthermore, SIO-UV was successfully deployed in real-world experiments with a remotely operated vehicle (ROV), demonstrating its outstanding performance in real-time processing, robustness, and accuracy.

Index Terms—Multiscale constant false alarm rate (MCFAR), reverse regression mapping method, simultaneous localization and mapping (SLAM), SIO-UV, underwater vehicles.

I. INTRODUCTION

STATE estimation and environmental perception are foundational for underwater vehicle mission execution. In terrestrial and aerial robotics, visual and LiDAR technologies [1] are commonly employed to achieve high-performance real-time simultaneous localization and mapping (SLAM). However, in underwater environments, strong absorption and scattering of light in water significantly shorten its propagation range,

limiting the application of visual and LiDAR technologies. Consequently, underwater vehicles primarily rely on acoustic sensors for perception. Among these sensors, forward-looking multibeam sonar (FLS) [2] stands out as a core sensor for underwater SLAM due to its cost-effectiveness and reliable imaging capabilities, providing stable environmental perception data. Therefore, this study focuses on developing SLAM methods for underwater vehicles based on FLS.

Over the past decade, FLS-based SLAM methods for underwater vehicles have achieved substantial progress. Among these methods, sonar image feature matching [3] and sonar planar point cloud matching [4] are currently the two core technologies widely applied to state estimation and map construction for underwater vehicles. Due to the sparsity of sonar data, these approaches typically rely on an inertial measurement unit (IMU) [5] to provide an initial state estimate, using external constraints to compensate for incomplete sonar data and prevent the vehicle from becoming isolated in terms of information. Despite the achievements of these methods, they also have notable limitations. First, underwater environments are prone to high noise and interference levels, reducing the reliability of sonar image feature matching in complex conditions [6]. Although these methods generally capture environmental features, background fluctuations around feature pixels at the pixel level make precise matching challenging with current feature matching operators. Second, due to the sparsity of 2-D point clouds, the performance of sonar planar point cloud matching is often sub-optimal in nonlinear optimization tasks [7]. Additionally, IMU-based state estimation commonly employs filtering methods. However, in long-range and large-scale mission scenarios, IMU errors tend to accumulate over time, reducing the accuracy of state estimation.

We propose a sonar-inertial odometry framework, SIO-UV, designed to address the localization and navigation challenges of underwater vehicles. We apply a smoothed linearization to the vertical projection model of the FLS and use a multiscale constant false alarm rate (MCFAR) denoising technique [8]. It also introduces an FLS reverse regression mapping method to convert sonar images into 3-D point clouds. The preintegration results from the raw IMU data are used as an initial pose estimate for the sonar point cloud, with the sonar odometry estimation further utilized to correct IMU biases in the overall factor graph. By incorporating a global factor graph for underwater vehicle state estimation, this framework effectively integrates FLS observations with IMU measurements, achieving joint optimization of the factor graph. The primary contributions of this work are as follows.

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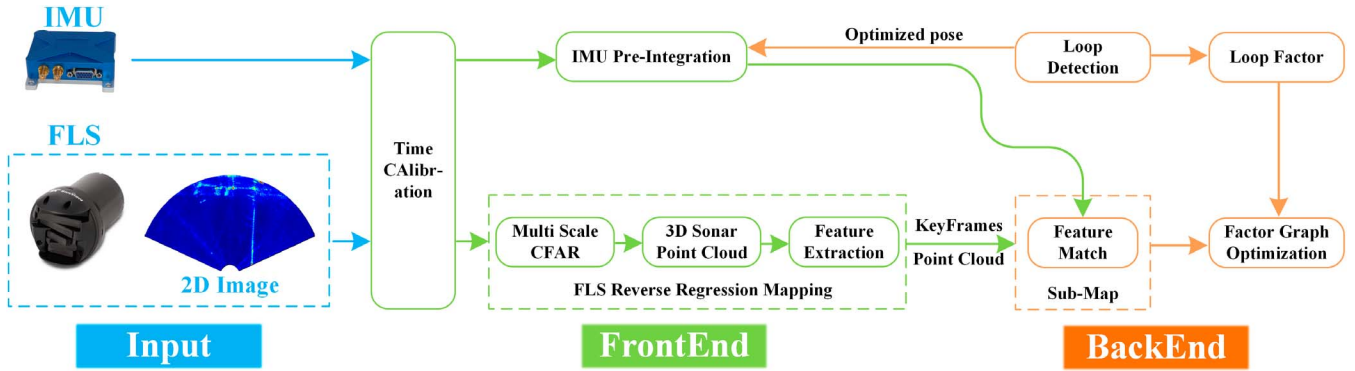


Fig. 1. Overall framework of SIO-UV. The FrontEnd processes include IMU preintegration and provide initial pose estimates for the BackEnd sonar odometry matching. Through the FLS reverse regression mapping algorithm, the FrontEnd converts 2-D images into 3-D sonar point clouds and performs feature extraction, thereby supplying raw data for the BackEnd sonar odometry matching. The BackEnd integrates the preintegrated poses with the sonar odometry poses through graph optimization, providing calibration offsets for the FrontEnd IMU preintegration.

- 1) We propose a rapid and robust sonar-inertial odometry framework based on a factor graph, designed for state estimation and environmental perception of underwater vehicles.
- 2) A robust FLS reverse regression mapping method is introduced, leveraging the MCFAR denoising algorithm, effectively converting sonar images into 3-D point clouds.
- 3) The proposed framework has been validated through extensive testing in both advanced simulation environments and real underwater vehicle deployments, demonstrating its robustness and efficiency.

II. RELATED WORK

Due to the rotational mechanism of the FLS transducer, its scanning range is limited in both the horizontal and vertical directions, resulting in sparse and uncertain environmental information. Consequently, it is often combined with an IMU to more accurately estimate the motion state of underwater vehicles. Next, we will discuss the two core fusion methods: filter and graph optimization.

A. Filter

In underwater vehicle SLAM, filter-based fusion of IMU and FLS data is a commonly used approach. The extended Kalman filter (EKF) and particle filter (PF) are two representative fusion methods. Bathymetric distributed particle SLAM (BPSLAM) [9] proposes an EKF-SLAM approach by integrating the motion model with sonar image feature matching results. However, this method may encounter significant errors in sonar image feature matching. Augmented extended Kalman filter SLAM (AEKF-SLAM) [10] stores vehicle poses and map landmarks in a single state vector, updating state parameters through recursive iterations. Additionally, scan-matching-based EKF methods [11] combine probabilistic scans and sonar range scans to effectively fuse data from different sensors. These EKF-based methods are sensitive to the complexity of the update process and linearization errors. The Rao-Blackwellized particle filter (RBPF) [12] uses sonar scan-matching results to estimate vehicle poses, with the advantage of building reliable occupancy grid maps [13].

Nevertheless, these methods are often based on sonar image feature matching. Once the matching results introduce substantial errors, the robustness of the entire system is compromised. It is a common issue in underwater environments.

B. Graph Optimization

Factor graphs exhibit significant advantages in underwater SLAM by clearly representing the relationships between state variables of the vehicle and sensor observation data as nodes and edges. The acoustic structure from motion (ASFM) [14] method integrates IMU and FLS data to construct a factor graph; however, its reliance on manually set positions for sonar image feature points undermines reliability. Another approach, RIAUL [15], employs A-KAZE features [16] for sonar image feature matching and establishes underwater landmarks. While it has shown some advantages, its applicability is limited to specific scenarios, making it unsuitable for larger underwater environments. These methods fail to fundamentally address the issue of significant errors in sonar image feature matching. The VMAE [17] method converts sonar images into 2-D point clouds and uses the iterative closest point (ICP) algorithm [18] for pose transformation between sonar frames, constructing sonar odometry. However, due to the sparsity of 2-D point clouds, the ICP matching results are often suboptimal. To enhance matching robustness, the method of supplementing sonar point cloud observations [19] utilizes a U-Net network [20] to map raw sparse 2-D point clouds into complete grid point clouds, significantly improving matching performance. However, this approach has the drawback of requiring prior grid calibration of the point clouds and is only applicable to specific scenarios. Additionally, converting sonar point clouds into a 2-D plane limits the effective construction of 3-D maps, thereby constraining its range of application.

III. RAPID AND ROBUST SONAR INERTIAL ODOMETRY

A. Framework Description

SIO-UV employs a classic SLAM framework divided into three main components: Input, FrontEnd, and BackEnd. This structure includes three threads, as shown in Fig. 1, consisting

of the IMU preintegration thread, the FLS reverse regression mapping thread, and the graph optimization thread.

The SIO-UV system requires IMU data and FLS images as two essential input types. First, timestamp calibration is performed on the IMU data and sonar images. Within the calibrated time range, the IMU data undergo preintegration to obtain the initial pose transformation T_{imu} . Next, FLS reverse regression mapping is conducted on the two corresponding sonar images, which includes MCFAR denoising, 3-D point cloud regression, and feature extraction. Specific operational details will be described in subsequent sections. After obtaining the sonar 3-D point clouds, key sonar frames are identified, and these key frames, along with the 3-D sonar point clouds and the pose transformation T_{imu} , are passed to the back end for further optimization. In the BackEnd module, feature matching is performed between the sonar key frames and the subgraph, using T_{imu} as the pose estimate, ultimately yielding the matched pose T_{sonar} for the sonar odometry.

As the number of key frames accumulates, factor graph optimization is performed over a sliding window, allowing the state estimation problem to be framed as a maximum a posteriori (MAP) estimation. Additionally, loop closure detection is conducted using the ICP algorithm. Once a loop is detected, the system performs a global optimization, distributing the overall error across each factor graph node. Ultimately, the optimized sonar odometry poses are fed back into the FrontEnd preintegration module to calibrate the IMU preintegration biases.

B. IMU Preintegration

IMU preintegration accumulates IMU data in a local coordinate system, avoiding repeated integration during nonlinear optimization, which reduces computational load and enhances data processing efficiency. During the IMU preintegration process for underwater vehicles, special attention must be given to the following motion states:

$$x = [P, R, V, b_a, b_g]^T. \quad (1)$$

Here, p represents the position variable of the vehicle, R denotes the rotation variable, V indicates the velocity variable, and b_a and b_g represent the biases in acceleration and angular velocity of the IMU, respectively.

After timestamp calibration, if the starting indices of the IMU data for the two sets of key sonar frames are i and j , the pose transformation T_{imu} between the two sets of sonar frames, according to the principles of IMU preintegration [21], can be expressed as

$$\begin{aligned} \Delta \tilde{P}_{ij} &= R_i^T \left(P_j - P_i - V_i \Delta t_{ij} - \frac{1}{2} g \Delta t_{ij}^2 \right) + \delta P_{ij} \\ \Delta \tilde{V}_{ij} &= R_i^T (V_j - V_i - g \Delta t_{ij}) + \delta V_{ij} \\ \Delta \tilde{R}_{ij} &= R_i^T R_j \text{Exp}(\delta \phi_{ij}). \end{aligned} \quad (2)$$

δP and δV represent the noise terms, while $\text{Exp}(\delta \phi)$ is the exponential map from Lie groups and Lie algebras, which converts the angular disturbance vector $\delta \phi$ into a small rotation matrix, indicating the small deviations introduced by noise.

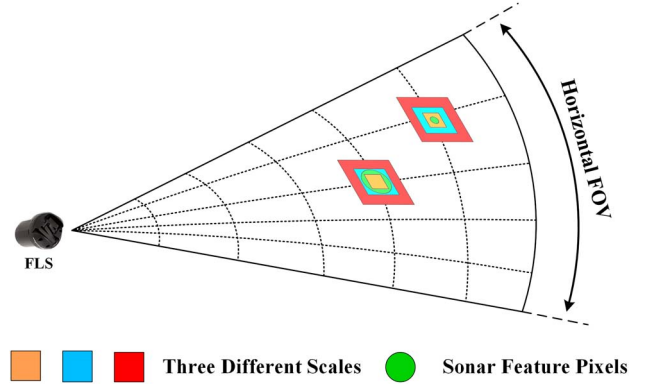


Fig. 2. Schematic diagram of MCFAR. Multiscale denoising is applied to feature pixels of different sizes, performing noise suppression and threshold calculation at each scale. The minimum value among the detection thresholds from all scales is selected as the final detection threshold. This strategy effectively ensures that feature miss-detection is avoided in complex noise backgrounds while enhancing the robustness and accuracy of sonar feature detection.

C. Reverse Regression Mapping Method for FLS

When new sonar frame data enters the SIO-UV framework, the FLS reverse regression mapping algorithm is first executed to convert the 2-D sonar images into 3-D sonar point clouds. This algorithm consists of several steps, including MCFAR denoising, 3-D point cloud smoothing mapping, and feature extraction.

In sonar image processing, noise is often nonhomogeneous and can be influenced by the complex underwater environment and sensor performance. Constant false alarm rate (CFAR) techniques [22] excel in suppressing noise and enhancing target detection. Based on this, we propose an MCFAR detection method, as shown in Fig. 2. MCFAR estimates noise at different scales and dynamically adjusts the detection threshold, enabling precise differentiation between target signals and noise signals, effectively addressing the challenges posed by multiple targets and complex noise backgrounds. For each pixel position, the received signal strength x follows the following probability density function:

$$f(x) = \frac{1}{\sigma} e^{-\frac{x}{\sigma}}, \quad x \geq 0. \quad (3)$$

Here, σ represents the average power of the local noise. In MCFAR, noise estimation is performed at multiple different scales. At each scale s , the noise level is estimated by selecting a background window of size N_s around the target pixel. Assuming the intensity values of the N_s pixels in the background window are $\{x_1, x_2, \dots, x_{N_s}\}$, the noise estimate $\hat{\sigma}_s$ at this scale is given by

$$\hat{\sigma}_s = \frac{1}{N_s} \sum_{i=1}^{N_s} x_i. \quad (4)$$

The size of the background window N_s increases with the scale s . Typically, multiple scales $s = 1, 2, 3$ are selected to estimate the noise values $\hat{\sigma}_s$ at each scale. Assuming the false alarm rate P_{fa} is predefined, we need to calculate the detection threshold T_s at each scale s based on the noise estimate $\hat{\sigma}_s$. The threshold

setting formula for each scale is given by

$$T_s = \alpha_s \hat{\sigma}_s. \quad (5)$$

α_s is the threshold factor, which depends on the predefined false alarm rate P_{fa} . At each scale s , the false alarm rate P_{fa} represents the probability that the signal strength x_{target} exceeds the detection threshold T_s

$$P_{fa} = \int_{T_s}^{\infty} \frac{1}{\sigma} e^{-\frac{x}{\sigma}} dx. \quad (6)$$

The integral result is calculated as follows:

$$P_{fa} = e^{-\frac{T_s}{\sigma}}. \quad (7)$$

Replacing T_s with $\alpha_s \hat{\sigma}_s$ and assuming $\hat{\sigma}_s = \sigma$, we can obtain the following:

$$P_{fa} = e^{-\alpha_s}. \quad (8)$$

Therefore, the threshold factor α_s for each scale s can be derived from the predefined false alarm rate P_{fa}

$$\alpha_s = -\ln(P_{fa}). \quad (9)$$

To enhance the robustness of detection, the final detection result can be achieved through a multiscale fusion approach, which combines the detection thresholds T_s from different scales to make more resilient decisions in more complex noise backgrounds. The multiscale fusion strategy we adopt is to take the minimum value of the thresholds across all scales as the final detection threshold T_{final}

$$T_{\text{final}} = \min_s T_s. \quad (10)$$

In FLS images, the noise patterns are diverse, and the noise or clutter characteristics may differ significantly on different scales. At smaller scales, images capture more detail but also tend to contain more tiny noise. At larger scales, the details of the noise may no longer be apparent but may mask some of the smaller targets. In the MCFAR method, the final threshold is calculated based on the minimum threshold of the three scales. The core idea of the strategy is as follows: at smaller scales, the threshold is low so that the system can be more sensitive to detect subtle anomalies, including tiny noises, thus ensuring that even smaller anomalies can be recognized; at larger scales, the threshold is high and is mainly used to suppress larger noises to ensure that the noises do not interfere with the results of the detections, which may miss some smaller targets. By choosing a minimum threshold, it can be ensured that even if the detected anomaly is relatively weak at one scale, it can be verified by other scales. This increases the robustness of the detection and reduces the risk of missed detections.

If only the smallest scale threshold is used, this may lead to over-sensitization, i.e., too strong a response to noise, and thus more false positives. Detection at small scales is more susceptible to noise and may incorrectly identify many irrelevant targets. Larger scales, on the other hand, help to suppress noise effectively; however, some smaller targets may be missed. Choosing the minimum threshold as a comprehensive judgment criterion can avoid too many false positives while ensuring that small

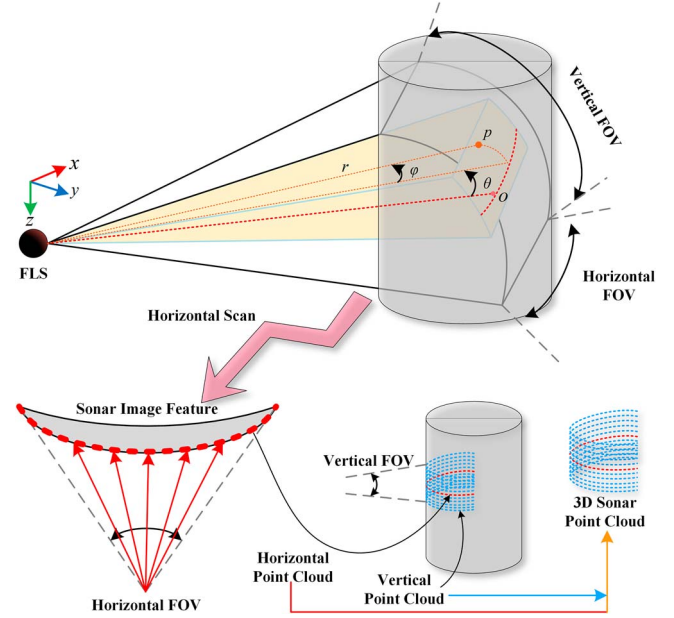


Fig. 3. Schematic diagram of converting the FLS projection model into a 3-D sonar point cloud. The sonar image is transformed into horizontal and vertical point clouds through horizontal and vertical scans based on the number of beams, collectively forming the 3-D sonar point cloud.

targets are not overlooked, thus achieving a more balanced and robust detection effect.

After the sonar image is processed using MCFAR denoising, the next step is to convert the image into a 3-D sonar point cloud, as illustrated in Fig. 3. In the FLS physical model, the detected points can be viewed as points in a spherical coordinate system $p[r, \theta, \varphi]$, where r is the beam range, θ is the azimuth angle, and φ is the elevation angle. To convert the points from the spherical coordinate system to points in the world Cartesian coordinate system $p[x, y, z]$, the following transformation equations can be used:

$$P = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = R \begin{pmatrix} \cos\theta\cos\varphi \\ \sin\theta\cos\varphi \\ \sin\varphi \end{pmatrix}. \quad (11)$$

In the horizontal opening angle direction, if the beam opening angle is α and the number of beams is h , the angular interval between beams is α/h . In the sonar image, simulated sonar wave detection is performed based on the initial position of the sonar's horizontal beam emission and the beam interval. When n pixel points are continuously detected along a beamline, the first detected pixel point is considered a valid point and is retained. This process is repeated for each beam, retaining the first point detected by each beam to obtain the point cloud features $\mathbb{F}_h$. This method differs from traditional point cloud conversion methods and resembles the scanning approach used in LiDAR systems [24]. Although retaining all sonar point clouds can provide more information, the presence of time-varying noise in sonar data means that keeping all data may reduce the robustness of interframe matching.

Since the FLS projects 3-D coordinates onto a 2-D plane through an unknown projection matrix, the elevation angle φ is

often neglected, making it difficult to directly recover the elevation information for each pixel from the sonar image. However, given that the elevation angle φ is typically small (generally between 10° and 20°), and with a vertical beam count of v , the angular interval between vertical beams is φ/v . This article assumes that there are pixel points on each beam, allowing for a scanning approach similar to that used in the horizontal direction, applied vertically. This process results in a stacking effect along the beam direction, generating vertical point cloud features $\mathbb{F}_v$. Ultimately, by combining the horizontal and vertical point cloud features, the desired 3-D sonar point cloud features $\mathbb{F}$ are generated.

The scheme vertically stretches the horizontal sonar point cloud by ignoring the effect of elevation angle in the vertical direction and extends it into a 3-D point cloud, which is reasonable in practical applications. Since the vertical opening angle of the sonar is small and the farthest vertical scanning distance is only about 1 meter, the influence of the vertical direction on the point cloud is small and can be effectively ignored. Meanwhile, the horizontal sonar has better resolution and lower point cloud density in the horizontal plane. Stretching it to the vertical direction can supplement the dimension of the point cloud, which can enhance the matching accuracy of the point cloud in the horizontal direction. The method has high feasibility and is especially suitable for the scenarios with scarce vertical data or large errors. By stretching and expanding the horizontal sonar image, the accurate information in the horizontal direction can be retained. It can provide a more complete 3-D point cloud structure in the subsequent processing of point cloud matching and feature extraction, thus improving the matching stability and processing efficiency.

Once the conversion of the 3-D sonar point cloud is completed, the next step is to perform feature extraction from the point cloud. It is important to note that the direct application of ICP or GICP for interframe matching [25] on these sparse 3-D sonar point clouds is inadvisable, as direct matching of sparse point clouds tends to be less robust, leading to significant errors and increased computational time. In this method, we first utilize the horizontal scanning point cloud features $\mathbb{F}_h$. Subsequently, following the LOAM approach [26], we extract corner features $\mathbb{F}_p$ and surface features $\mathbb{F}_s$ from the 3-D sonar point cloud. The extraction criteria are based on the curvature of the points: points with a curvature greater than a specified threshold are designated as corner features, while those with a curvature below this threshold are designated as surface features. Thus, we have successfully completed the conversion from 2-D sonar images to 3-D sonar point clouds and extracted three types of point cloud features.

The stretched point cloud is supplemented with data in the vertical direction by assumptions that lack true vertical information but are able to retain enough horizontal point cloud for matching and processing. This simplified processing does not imply a flaw in the scheme but rather is a reasonable assumption based on measurement accuracy limitations in real-world environments and is intended to reduce computational complexity and improve efficiency. Although the stretched point cloud may contain mainly features such as edges and corner points, this can

still effectively support subsequent matching and recognition tasks in some cases.

For feature extraction, although the stretched point cloud may be dominated by edges and corner points, these features are still very important in many applications, especially in scenes where fast matching and object recognition are required.

D. Factor Graph Optimization

The factor graph of this framework is illustrated in Fig. 4. This factor graph contains only one type of node: the underwater vehicle pose x_i , along with three types of constraint factors: the IMU preintegration factor u_i , the sonar odometry factor s_i , and the loop closure factor l_i (not explicitly marked in the figure). The goal of factor graph optimization is to find the maximum likelihood set of vehicle poses $X = \{x_i\}$ given all observations $Z = \{u_i, s_i, l_i\}$. The optimal set satisfying this condition is denoted as X^*

$$\begin{aligned} X^* &= \underset{x}{\operatorname{argmax}} p(X|Z) \\ &= \underset{x}{\operatorname{argmax}} p(x_0) \prod_{k=1}^m p(u_k | x_{k-1}, x_k) \\ &\quad * \prod_{i=1}^n p(s_i | x_{i-1}, x_i) \prod_{j=1}^l p(l_j | x_*, x_{**}). \end{aligned} \quad (12)$$

We apply Bayes' theorem to obtain the MAP probability of X^* . Assuming that both the IMU and sonar measurements follow a Gaussian noise model, the observation model can be expressed as

$$f(x_{i-1}, x_i) + \mathcal{N}(0, Q_i) \quad (13)$$

$$g(x_{i-1}, x_i) + \mathcal{N}(0, R_i) \quad (14)$$

where $f(x_{i-1}, x_i)$ and $g(x_{i-1}, x_i)$ represent the IMU preintegration and sonar odometry measurements between x_{i-1} and x_i , respectively. The loop closure model also employs a Gaussian noise model, expressed as $h(x_{i_k}, x_{i_m}) + \mathcal{N}(0, U_i)$. Under the Gaussian assumption, the MAP probability problem can be simplified to a nonlinear least squares problem, formulated as

$$\begin{aligned} X^* &= \underset{x}{\operatorname{argmin}} \left[-\log_{p(x_0)} \prod_{k=1}^m p(u_k | x_{k-1}, x_k) \right. \\ &\quad \left. * \prod_{i=1}^n p(s_i | x_{i-1}, x_i) \prod_{j=1}^l p(l_j | x_*, x_{**}) \right] \\ &= \underset{x}{\operatorname{argmin}} \left[\|x_0\|_Q^2 + \sum_{k=1}^m \|f(x_{k-1}, x_k) - u_k\|_Q^2 \right. \\ &\quad \left. + \sum_{i=1}^n \|g(x_{i-1}, x_i) - s_i\|_R^2 + \sum_{j=1}^l \|h(x_*, x_{**}) - l_j\|_U^2 \right]. \end{aligned} \quad (15)$$

Subsequently, the Gaussian-Newton method or the Levenberg-Marquardt algorithm is employed for nonlinear optimization to find an approximate solution that approaches the minimum. For detailed information on the solution process, please refer to iSAM2 [27].

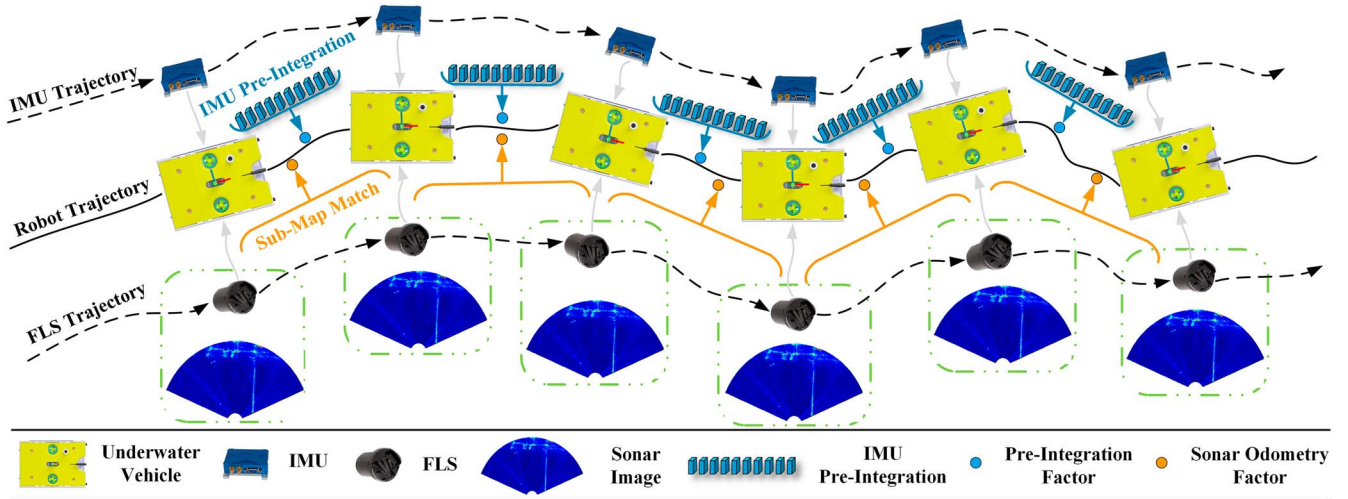


Fig. 4. Overall framework of factor graph optimization.

The sonar odometry factor is derived from the three types of sonar point cloud features obtained by the front end. First, for the horizontal scanning point cloud feature $\mathbb{F}_h$, which resembles a 2-D lidar scanning feature, we utilize PL-ICP [28] to compute the transformation matrix T_h . This method is particularly suitable for line structures and exhibits greater robustness against noise. Next, for the corner features $\mathbb{F}_p$ and surface features $\mathbb{F}_s$, we employ the point-surface matching approach from LIO-SAM [29] for pose matching of the 3-D sonar point clouds, resulting in the transformation matrix T_c . It is important to note that the matching process involves finding correspondences between the current sonar frame and the relevant features in the local map, categorized as SubMap-Match, with the initial estimate provided by the IMU preintegration result T_{imu} . Finally, we fuse T_h and T_c using a weight w to obtain the final sonar odometry pose transformation matrix T_{sonar}

$$T_{sonar} = w_1 * T_h + w_2 * T_c. \quad (16)$$

In the processing of loop closure factors, the SIO-UV system employs a distance space search method to detect potential loop closures by comparing the pose distances between historical keyframes and the current keyframe. When processing each new keyframe, the system calculates the Euclidean distance to all historical keyframes. If the distance between a historical frame and the current frame is below a predefined threshold, the historical frame is considered a loop closure candidate. Once potential loop closure candidates are identified, the system executes an ICP-based alignment process for the sonar point clouds. By performing ICP matching between the point cloud data of the current frame and the candidate frame, if the matching error falls within a specified threshold, the loop closure detection is deemed successful. Upon confirming the existence of a loop closure, the system generates the corresponding loop closure factor and adds it to the factor graph. This loop closure factor acts as a new constraint that connects the current frame to the historical frame where the loop closure occurs, thereby reducing accumulated errors. By incorporating this constraint into the factor graph optimization, the system can further refine the overall

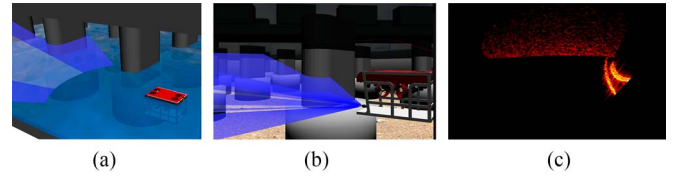


Fig. 5. Simulation of the underwater environment: (a) Rexrov model; (b) BlueView M900 sonar model; and (c) sonar imaging results.

trajectory estimate, enhancing global consistency and localization accuracy.

IV. EXPERIMENTS AND RESULTS

A. Simulation Experiment

In this study, an underwater environment is constructed on the ROS Noetic simulation platform using the Gazebo simulation tool, as illustrated in Fig. 5. The type of underwater vehicle used is a remotely operated vehicle (ROV), specifically employing the Rexrov model [30] from the UUV simulator platform. This model is equipped with an IMU sensor. Additionally, to implement the FLS functionality, the Blueview M900 sonar model [31] from the Dave simulation platform is introduced. The Dave platform, developed based on UUV simulator, offers good integration compatibility. The advantages of this model include support for CUDA-based GPU-accelerated raster simulation, which accurately simulates the phase and reflection characteristics of acoustic signals while employing a point scattering model to provide raw sonar intensity range data, ultimately generating imaging results highly similar to real sonar. Notably, the FLS model features a horizontal field of view of 130° , a vertical field of view of 15° , and a maximum scanning distance of 10 m, with 512 horizontal beams and 256 vertical beams. It is important to note that the Rexrov operates at relatively low speeds in the underwater environment.

The simulation experiment set up two underwater environments, as shown in Fig. 6. To evaluate the localization and mapping performance of the SIO-UV framework under low visibility

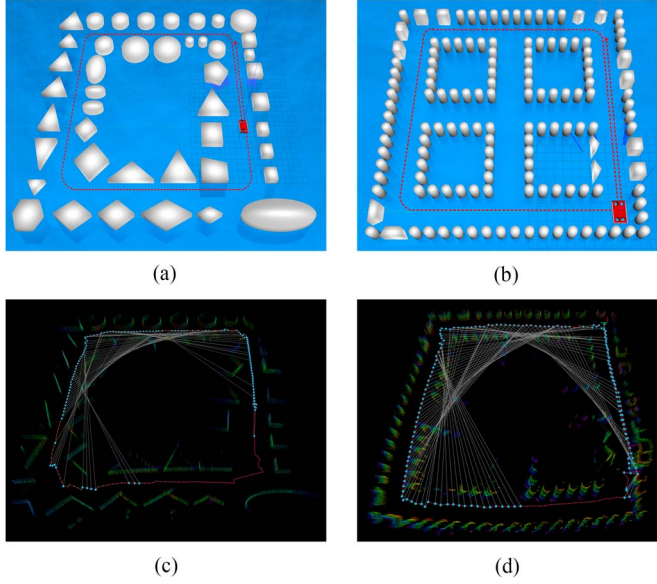


Fig. 6. Two underwater scenarios and the corresponding map construction results: (a) high distinguishability scenario 1; (b) low distinguishability degraded scenario 2; (c) sonar point cloud map for scenario 1; and (d) sonar point cloud map for scenario 2.

conditions, two scenarios were designed: scene 1, measuring $40 \times 40 \text{ m}^2$ with distinct features, and scene 2, measuring $30 \times 30 \text{ m}^2$ with low distinguishability and degraded characteristics. These scenarios allow for the verification of whether the SIO-UV framework can successfully achieve loop closure detection and construct sonar point cloud maps in different environments. As seen in Fig. 6, the SIO-UV framework successfully generated 3-D sonar point cloud maps in both scenarios, while other methods were only able to produce 2-D point cloud maps. Furthermore, the constructed point cloud maps matched the original map's dimensions and shape. In the low distinguishability scene 2, the degraded characteristics led to some deviations in localization when returning to the starting point. Nevertheless, loop closure detection was successfully triggered, and a degree of alignment between the point cloud was achieved during the factor graph optimization process. This success can be attributed to the SIO-UV framework's ability to build robust 3-D sonar point clouds and its strong capabilities in sonar odometry pose matching. However, the limited field of view reduced the confidence in the matching results, resulting in decreased constraint accuracy in the factor graph, which ultimately affected the optimization outcome.

As shown in Fig. 7, we compared the trajectories of three methods against the actual trajectory of the ROV, with the starting position set at the origin. Bruce-SLAM [17] estimates the IMU state based on the EKF and utilizes 2-D sonar point clouds for odometry estimation. However, because it does not focus specifically on edge features and extracts all sonar point cloud data, significant sonar noise increases during ROV turns, exacerbating overall state estimation errors and adversely affecting system robustness. ASFM-SLAM [14] employs the A-KAZE operator for feature matching in sonar images, but it experiences a high rate of mismatches during the matching process, preventing the system from running smoothly. This issue is particularly

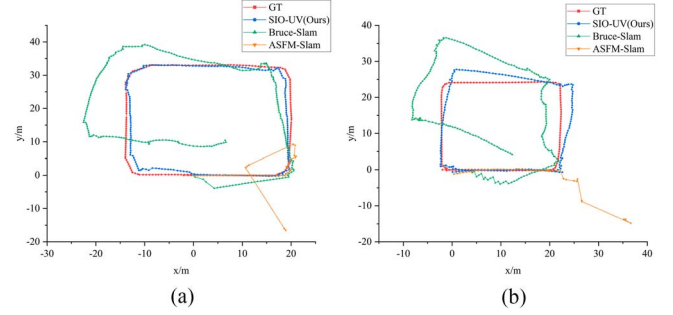


Fig. 7. Trajectories in the two underwater scenarios. The compared trajectories include the actual ground truth (GT), the proposed SIO-UV trajectory, the Bruce-SLAM method trajectory, and the ASFM-SLAM trajectory.

TABLE I
EVALUATION RESULTS OF UNDERWATER SCENE EXPERIMENTS

Scene	Time (s)	Evaluation	Methods		
			SIO-UV(Ours)	Bruce-SLAM	ASFM-SLAM
1	232.48	Mean APE (m)	1.729	7.152	10.315
		RMSE (m)	1.794	8.480	15.311
2	213.27	Mean APE (m)	1.705	8.203	13.776
		RMSE (m)	1.862	9.300	16.108

pronounced during initial turns, leading to significant drift. In contrast, the proposed SIO-UV framework integrates IMU pre-integration and estimates odometry based on 3-D sonar point clouds. Even in low distinguishability degraded scenarios, it achieves accurate state estimation, demonstrating significantly better robustness than the other two methods.

Furthermore, as shown in Table I, we evaluated the accuracy of each system using the average absolute pose error (APE) and root mean square error (RMSE) metrics. A key distinction of the SIO-UV framework is its incorporation of IMU preintegration to provide an initial pose estimate, combined with 3-D sonar point clouds. The results indicate that these enhancements significantly improve the robustness of odometry matching, with final accuracy outperforming traditional methods based on sonar image feature matching and 2-D sonar point cloud matching. Methods relying solely on image feature matching performed the worst, exhibiting substantial matching errors, indicating their unsuitability for sonar environments.

B. MCFAR Analysis Discussion

In this section, the advantages and disadvantages of MCFAR and the smallest of cell average CFAR (SOCA-CFAR) will be analyzed in depth and discussed with experimental results.

MCFAR employs multiscale processing to perform CFAR detection at multiple scales and selects the smallest threshold, thus enhancing the adaptability to targets of different sizes and improving the robustness to complex backgrounds. This approach performs particularly well in cases where the target size varies greatly or the background noise is inhomogeneous.

TABLE II
COMPARISON OF DIFFERENT DENOISING METHODS

Methods	Scale	Windows	Scenario	Time/ms	ICP Matching Error
CFAR	1	5	Simulation 1	5	29.07%
			Simulation 2		35.21%
MCFAR	3	5/7/9	Simulation 1	11	18.56%
			Simulation 2		21.37%
SOCA-CFAR	1	5	Simulation 1	6	23.19%
			Simulation 2		26.42%

Meanwhile, SOCA-CFAR dynamically adapts to the characteristics of different targets by adaptively adjusting the size of the detection window. The noise distribution is computed from the local environment, allowing it to be optimized for specific background noise. Overall, MCFAR puts more emphasis on multi-scale processing to enhance detection sensitivity, while SOCA-CFAR focuses on adaptively adjusting the window to optimize the detection performance in specific backgrounds.

Since sonar image denoising is an unsupervised problem and it is not possible to measure the superiority of the algorithm directly by the predefined denoised images, this article uses the percentage of matching error between all frames for comparison. The matching method is unified by the ICP algorithm, but the initial IMU position estimation is not provided to avoid the influence of external information on the experimental results. The final ICP matching error percentage is calculated from the average of displacement error percentage and rotation error percentage. Since this article focuses on the exploration mission of the underwater vehicle in the same horizontal plane, the rotational error is calculated only for the YAW angular error, ignoring other rotational degrees of freedom.

The sonar image scale of 256×256 is used for the experiments, and the window parameter settings of single-scale CFAR, MCFAR, and SOCA-CFAR are shown in Table II. During the computation process, all methods are accelerated by CUDA parallel acceleration to improve the computational efficiency. Thanks to the parallel computation, the overall computation time can meet the requirements. Since MCFAR needs to perform CFAR detection on multiple scales, the computational complexity is higher, and its processing time is about twice as much as that of SOCA-CFAR, but the overall running time is still short enough to meet the requirements of real-time applications. In simulation scenario 1, the matching accuracy of MCFAR is better than SOCA-CFAR by about 5%, and compared with the traditional single-scale CFAR method, the matching accuracy of MCFAR is improved by 12%. In simulation scenario 2, the lack of differentiation of the background features causes the ICP to perform poorly under sparse point cloud conditions. In addition, the limitation that the horizontal field of view of the sonar is only 130° which leads to a larger matching error than in simulation scenario 1.

The experimental results show that MCFAR improves the matching accuracy through multiscale processing, and has better robustness and adaptability than single-scale CFAR and SOCA-CFAR. Although the computation time is relatively long, it

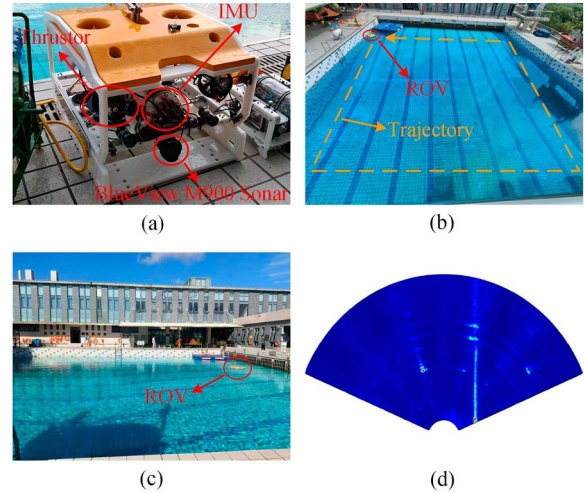


Fig. 8. ROV deployment in a real pool scenario: (a) ROV equipped with IMU and M900 sonar; (b) panoramic view of the pool with predefined trajectory; (c) ROV operating in the pool; and (d) M900 sonar imaging results.

shows obvious advantages in terms of denoising effect and final matching error. In particular, MCFAR has better detection and matching performance in scenarios with large changes in target size or complex background noise.

C. Real Experiment

To validate the performance of the SIO-UV framework under real underwater robot deployment, experiments were conducted in a $25 \times 25 \text{ m}^2$ pool, as shown in Fig. 8. For convenience, an ROV equipped with a BlueView M900 sonar and a MEMS inertial navigation module (providing IMU data) was used. Blue-View M900 is a 900–2250 dual-frequency multibeam imaging sonar, which has a detection angle of 130° and a range of 10–100 m. In addition, sensor data acquisition was performed using a Raspberry Pi unit inside the ROV connected to a host computer via a communication cable, with a ROS system deployed for the master and slave. Note that the sonar is mounted under the front of the ROV, while the IMU is located inside the control compartment, in the center of the ROV. During the experiment, the ROV was manually operated along a predefined trajectory, ensuring it remained on the water's surface while moving at a low speed to avoid submerging. Throughout the experiment, sensor data were collected, with the FLS sampling frequency set to 5 Hz and the IMU sampling frequency at 100 Hz, maintaining consistency with the simulation experiments.

Fig. 9 presents the results of the pool experiment. Due to sonar image noise and the presence of protective railings from the wave-making devices at the pool's bottom, the generated sonar point cloud map exhibits some minor gaps. However, the overall shape and dimensions of the map are consistent with the pool's design. Regarding the operating trajectories, our proposed framework shows slight drift during turns. Nonetheless, due to the final loop closure processing, the overall trajectory aligns closely with the true planned path. In contrast, the Bruce-SLAM and ASFM-SLAM methods experience significant errors in 2-D point cloud matching and image feature matching due to

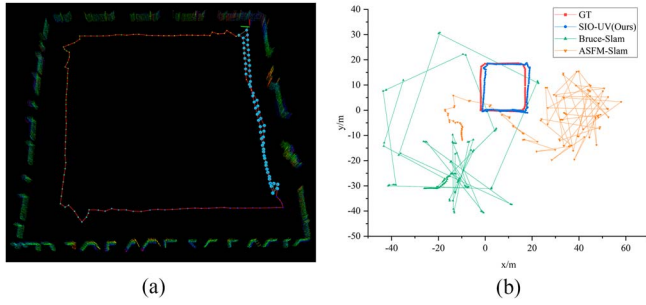


Fig. 9. Experimental results of the ROV deployment in the pool: (a) sonar point cloud map of the pool constructed by the SIO-UV framework; and (b) operating trajectories of different methods.

TABLE III
EVALUATION RESULTS OF POOL SCENE EXPERIMENTS

Scene	Evaluation	Methods		
		SIO-UV (Ours)	Bruce-Slam	ASFM-Slam
Pool	Mean APE (m)	1.623	16.364	23.936
	RMSE (m)	1.677	22.200	27.748

the high similarity of sonar images in the pool environment. Additionally, the EKF-based IMU estimation introduces cumulative errors, making it difficult for these two methods to achieve accurate state estimation for the ROV in degraded real-world scenarios. Table III provides detailed comparative metrics. The results indicate that the SIO-UV framework achieves a localization error of approximately 1.6 m in real-world settings, which is comparable to the simulation results and significantly outperforms the other two methods, demonstrating a clear advantage in system robustness.

D. Feature Extraction Analysis Discussion

In this section, the feature extraction approaches of the methods in this article, Bruce's method and ASFM method, are analyzed and discussed to evaluate their advantages and disadvantages in different scenarios. All methods use the SIO-UV frame, with the difference that the Bruce method uses SOCA-CFAR sonar image denoising and combines it with 2-D point cloud ICP matching, while the ASFM method uses A-KAZE sonar image feature extraction and position matching. The evaluation metrics use the percentage of position matching error over all frames, and the size of the sonar image is 256×256 . The final experimental results are shown in Fig. 10.

Bruce removes noise from SOCA-CFAR sonar images and handles feature extraction using 2-D point cloud ICP matching. This method performs better in noise removal and can effectively reduce the impact of image noise on feature extraction. However, 2-D point cloud ICP matching is constrained by the complexity of the underwater environment and the resolution of the sonar image, especially the limitation of the sonar's field of view, which leads to a larger matching error when the navigator turns. ASFM method uses the A-KAZE sonar image feature

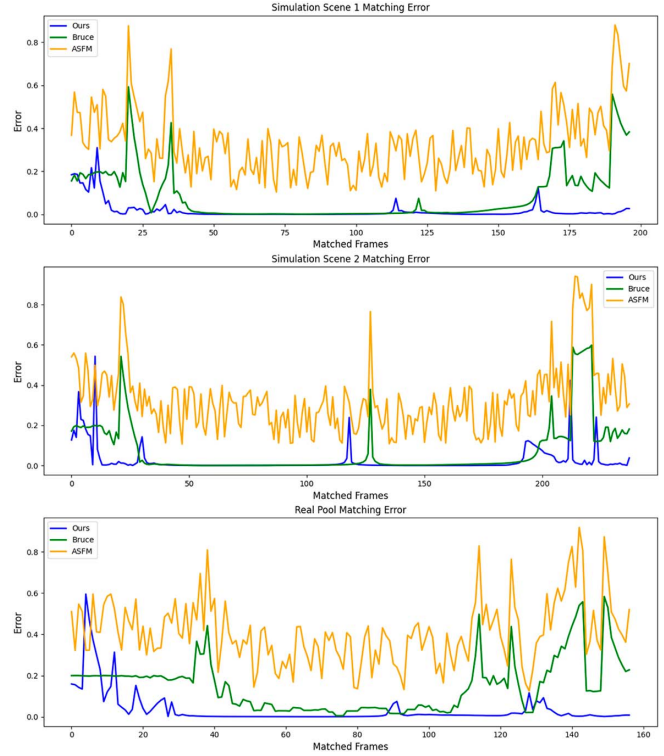


Fig. 10. Results of error comparison of different feature extraction methods.

extraction and performs the position matching, which relies on the key feature points. However, sonar images in underwater environments are often disturbed by noise and have time-varying characteristics, which may make the features extracted by the A-KAZE algorithm unstable or inaccurate. As a result, the matching error of the ASFM method is large in the experiments. In contrast, the method in this article has higher accuracy and robustness in feature extraction and can effectively extract feature points that are helpful for localization in different scenarios. Even in the scenario where the navigator is turning, limited by the field of view of the sonar, the method in this article can still realize very small matching error.

E. Time Efficiency

Table IV presents the runtime of our proposed SIO-UV framework. Based on the performance of the FrontEnd and BackEnd threads, all methods can meet the real-time performance requirements for an IMU frequency of 100 Hz and an FLS frequency of 5 Hz. This efficiency is attributed to the sparsity of factor graph nodes and sonar data, which significantly reduces the optimization time. Additionally, the runtime of the SIO-UV method is approximately 5 ms longer than that of the other two methods on average. However, it demonstrates superior accuracy and robustness, making this additional time cost acceptable. In the FLS reverse regression mapping algorithm of SIO-UV, the FrontEnd computation is relatively time-consuming due to the presence of MCFAR denoising. We employed convolution optimization techniques [32] and utilized CUDA acceleration to keep the FrontEnd runtime within an acceptable range. Without

TABLE IV
TIME EFFICIENCY COMPARISON

Scene	Stage(ms)	Methods		
		SIO-UV (Ours)	Bruce- Slam	ASFM- Slam
Simulation Scenario 1	FrontEnd	15	10	19
	BackEnd	50	50	50
Simulation Scenario 2	FrontEnd	17	12	20
	BackEnd	52	51	50
Pool Scenario	FrontEnd	22	15	23
	BackEnd	60	60	60

convolution optimization, the FrontEnd runtime is approximately 50 ms, while convolution optimization achieves about a threefold speedup.

V. CONCLUSION AND FUTURE WORK

We proposed a rapid and robust sonar-inertial odometry framework named SIO-UV, designed for state estimation and map construction of underwater vehicles in complex environments. SIO-UV builds factor graph nodes by integrating IMU and FLS observation data, using IMU preintegration to estimate the motion trajectory of the underwater vehicle and provide initial pose optimization for sonar observations. To enhance the robustness of interframe matching, we introduced an FLS reverse regression mapping method that converts FLS imaging data into 3-D sonar point clouds, significantly improving the reliability of interframe matching. Comprehensive performance evaluations were conducted in both advanced simulation environments and real-world ROV deployment experiments in a swimming pool. The results indicate that SIO-UV achieves a significant improvement in accuracy compared to other methods, maintaining errors within a small range in underwater environments with no prior information. Future work will focus on integrating underwater vision and ultrashort baseline (USBL) technology to generate more complete 3-D sonar point clouds, aiming to reduce cumulative drift in sparse underwater feature environments.

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